Position Generalization in Translated FashionMNIST

Architecture, patch scale, and patch embedding

Final English report | Controlled experiments · seed 42

Abstract. We test whether image classifiers retain accuracy when object position changes between training and testing. CNN is the strongest baseline and reaches 35.40% on the hardest fixed-to-random transfer. Within the ViT family, patch size 8 gives the best accuracy-cost balance. Conv2d and Flatten + Linear patch embeddings remain within 0.61 percentage points.

1 Question

A classifier may learn class appearance and object location together. This study isolates the effect of location by comparing random-position images (A) with centered images (B).

Setting	Training	Testing	Purpose
A -> A	random	random	in-distribution reference
B -> B	center	center	fixed-position reference
A -> B	random	center	random-to-fixed transfer
B -> A	center	random	hardest position shift

2 Controlled protocol

Each configuration trains one model on A and one on B. Validation chooses the checkpoint; the official test set is used once for final evaluation.

Item	Controlled choice
Data	FashionMNIST on a 64 x 64 black canvas
Split	90% train, 10% validation; official test set held out
Training	15 epochs, AdamW, seed 42
Selection	best validation checkpoint
Reporting	test accuracy and elapsed training time

Primary measure. B -> A is the hardest setting because a model trained only on centered objects must classify objects across the canvas.

Architecture and Position Shift

MLP, CNN, and patch-16 ViT under one evaluation protocol

3 Accuracy

Configuration	Parameters	A -> A	B -> B	A -> B	B -> A
MLP	542,474	71.89	89.57	71.97	13.99
CNN	205,994	92.35	93.14	92.87	35.40
ViT, patch 16	829,834	80.07	88.87	80.97	16.91

Table 1. Test accuracy (%). The best value in every column belongs to CNN.

Accuracy on fixed-center train -> random-position test

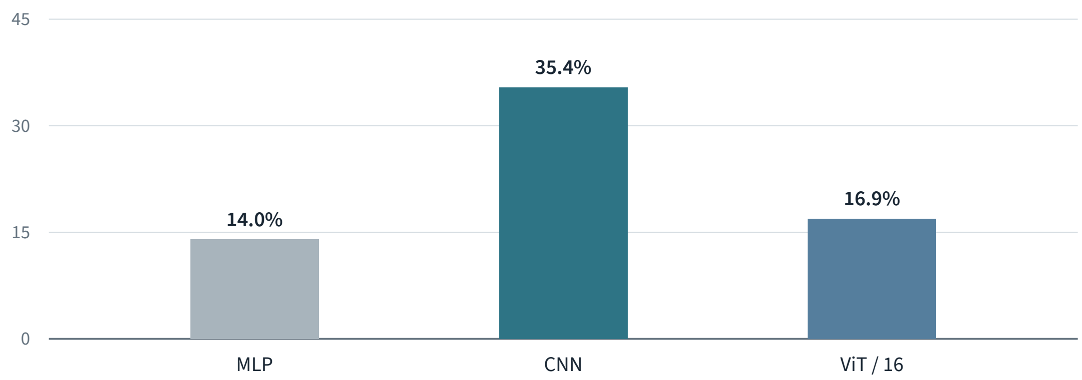

Figure 1. CNN retains substantially more accuracy on B -> A.

BEST B -> A	CNN ADVANTAGE	SMALLEST MODEL
35.40%	+18.49 pp	205,994
CNN	over patch-16 ViT	CNN parameters

4 Effect of the position shift

Fixed-position accuracy before and after the test distribution shifts

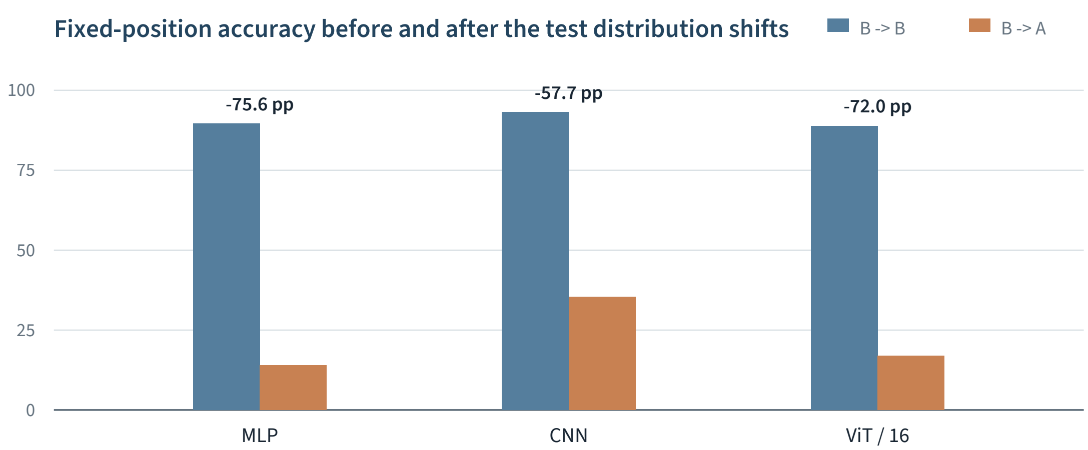

Figure 2. All models degrade after centered training is tested at random positions.

Patch Scale and Embedding

Two controlled changes to the shared Transformer configuration

5 Patch scale

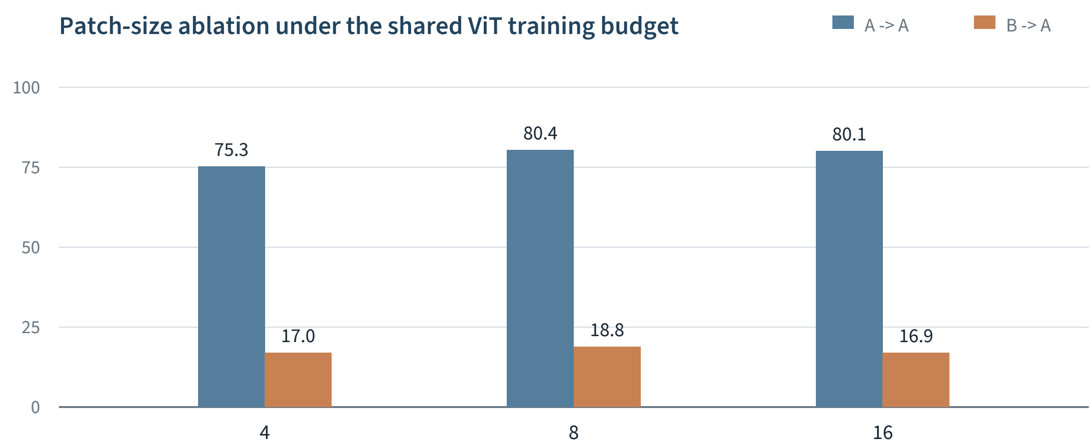

Figure 3. Patch size 8 gives the strongest observed trade-off.

Patch	Tokens	A -> A	B -> B	A -> B	B -> A	Time
4	256	75.28	87.22	74.97	17.02	8.93 min
8	64	80.40	88.27	81.13	18.81	7.36 min
16	16	80.07	88.87	80.97	16.91	6.46 min

Table 2. Accuracy (%) and total time for the A- and B-trained models.

Result. Patch size 8 leads on A -> A, A -> B, and B -> A. Patch size 4 raises the token count to 256 and is slower without an accuracy gain.

6 Patch embedding

For non-overlapping patches, a strided Conv2d and a shared linear layer perform the same type of patch-wise projection. Their results remain close.

Test	Conv2d	Flatten + Linear	Absolute gap
A -> A	80.07	79.67	0.40
B -> B	88.87	88.80	0.07
A -> B	80.97	80.48	0.49
B -> A	16.91	16.30	0.61

Table 3. Test accuracy (%) for patch size 16. Maximum absolute gap: 0.61 points.

DISCUSSION

Conclusions and Reproducibility

What the controlled comparisons support, and what they do not

7 Conclusions

01 **Architecture dominates.** CNN leads all four settings and retains the most accuracy under fixed-to-random transfer.

02 **Patch size 8 is the practical ViT choice.** It gives the best B -> A result without the 256-token cost of patch size 4.

03 **Embedding syntax is not decisive.** Conv2d and Flatten + Linear differ by at most 0.61 percentage points.

8 Limitations

Scope

- One formal seed; no variance estimate.
- Model sizes are not exactly matched.
- Runtime is hardware-specific.

Interpretation rule

Differences below 1 percentage point are treated as practically similar. Results support a controlled course comparison, not a population-level claim.

9 Reproducibility

Check	Recorded value
Code	translated_fashionmnist/experiments
Metrics	results/metrics.csv
Seed / epochs	42 / 15
Environment	Python 3.12, PyTorch 2.11, CUDA 12.8
Hardware	NVIDIA GeForce RTX 5070 Laptop GPU

```
python -m translated_fashionmnist.experiments.compare --groups all --download
```

> **External record.** The teammate repository is retained only as provenance. Its reported best values use a different checkpoint-selection procedure, so they are not treated as a statistically equivalent baseline.

10 References

[1] H. Xiao, K. Rasul, and R. Vollgraf. Fashion-MNIST: a Novel Image Dataset for Benchmarking Machine Learning Algorithms. arXiv:1708.07747, 2017.

[2] A. Dosovitskiy et al. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. ICLR, 2021.

[3] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner. Gradient-Based Learning Applied to Document Recognition. Proceedings of the IEEE, 1998.

[4] I. Loshchilov and F. Hutter. Decoupled Weight Decay Regularization. ICLR, 2019.

[5] kicious. translated-fashion-mnist-vit, recorded commit 943fa7b. https://github.com/kicious/translated-fashion-mnist-vit